

Ex.no:2	EDA-DATA IMPORT AND EXPORT

Aim:

To inspect dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read Food_Waste_Output.csv using `pd.read_csv()`.
3. Inspect the data structure using `.head()`, `.tail()`, `.info()`, `.describe()`, and `.isnull().sum()`.
4. Export the DataFrame to .csv and .xlsx formats using `.to_csv()` and `.to_excel()`.
5. Connect to food_waste.db using sqlite3 and write the data into an SQL table using `.to_sql()`.
6. Close the database connection and end execution.

Program:

```
import os
```

```
import pandas as pd
```

```
import sqlite3
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
df = pd.read_csv("E Commerce.csv")
```

```
print("First 5 records:")
```

```
print(df.head())
```

```
print("\nLast 5 records:")
```

```
print(df.tail())
```

```
print("\nDataset information:")
```

```
df.info()
```

```
print("\nDataset shape:")
```

```
print(df.shape)
```

```
print("\nColumns:")
```

```
print(df.columns.tolist())
```

```
print("\nData types:")
```

```
print(df.dtypes)
```

```
print("\nStatistical summary:")
```

```
print(df.describe(include="all"))
```

```
print("\nMissing values:")
```

```
print(df.isnull().sum())
```

```
df.to_csv("e_commerce_data.csv", index=False)
```

```
df.to_excel("e_commerce_data.xlsx", index=False)
```

```
conn = sqlite3.connect("e_commerce.db")
```

```
df.to_sql("e_commerce", conn, if_exists="replace", index=False)
```

```
conn.close()
```

```
print("\nCSV, Excel and SQLite files created successfully.")
```

Output:

```
Choose Files E Commerce.csv
... E Commerce.csv(text/csv) - 241227 bytes, last modified: 8/15/2026 - 100% done
Saving E Commerce.csv to E Commerce.csv

  order_id customer_id age product_id country signup_date \
0  ORD5000   CUST1000   39   PROD200   Canada  01-07-2021
1  ORD5001   CUST1001   61   PROD201     USA  10/19/2020
2  ORD5002   CUST1002   26   PROD202  Pakistan  06-10-2023
3  ORD5003   CUST1003   54   PROD203    India   7/30/2023
4  ORD5004   CUST1004   50   PROD204    India  12-09-2020

  last_purchase_date cancellations_count subscription_status order_date \
0          2/21/2023                0             active   8/20/2024
1          12-08-2021                0             active   7/17/2025
2          09-04-2023                3          cancelled  03-12-2025
3          2/20/2024                4             paused   9/19/2024
4          9/14/2024                0             active   08-08-2024

  unit_price quantity purchase_frequency preferred_category product_name \
0        78.21         5                37             Sports   Football
1        64.02         8                35             Sports  Refrigerator
2       604.14         2                44          Electronics   Hoodie
3        84.66         4                 1             Sports   Conditioner
4        62.66         7                35             Sports   Smartwatch

Statistical summary:
  order_id customer_id age product_id country signup_date \
count      2000      2000 2000.000000      2000      2000      2000
unique      2000      2000         NaN      2000         6      926
top    ORD6983   CUST2983         NaN   PROD2183   Germany  04-10-2023
freq         1         1         NaN         1        360         8
mean         NaN         NaN  44.080500         NaN         NaN         NaN
std          NaN         NaN  14.798091         NaN         NaN         NaN
min          NaN         NaN  18.000000         NaN         NaN         NaN
25%          NaN         NaN  31.000000         NaN         NaN         NaN
50%          NaN         NaN  45.000000         NaN         NaN         NaN
75%          NaN         NaN  57.000000         NaN         NaN         NaN
max          NaN         NaN  69.000000         NaN         NaN         NaN
```

```

...   count    unit_price    quantity    purchase_frequency    preferred_category \
      unique         NaN         NaN         NaN         5
      top         NaN         NaN         NaN    Clothing
      freq         NaN         NaN         NaN
      mean    324.719460    4.948500    25.150000         NaN
      std     361.548405    2.557757    14.158752         NaN
      min       2.850000    1.000000    1.000000         NaN
      25%      111.127500    3.000000    13.000000         NaN
      50%      206.515000    5.000000    25.000000         NaN
      75%      384.507500    7.000000    37.000000         NaN
      max     1991.630000    9.000000    49.000000         NaN

      product_name    category    gender
count             2000             2000             2000
unique              40              5              3
top      Sneakers    Clothing    Female
freq              72             426            1015
mean             NaN             NaN             NaN
std              NaN             NaN             NaN
min              NaN             NaN             NaN
25%              NaN             NaN             NaN
50%              NaN             NaN             NaN
75%              NaN             NaN             NaN

```

Missing values:

```

order_id          0
customer_id       0
age              0
product_id        0
country           0
signup_date       0
last_purchase_date 0
cancellations_count 0
subscription_status 0
order_date        0
unit_price        0
quantity          0
purchase_frequency 0
preferred_category 0
product_name      0
category          0
gender            0
dtype: int64

```

CSV, Excel and SQLite files created successfully.

Result: